THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 2: **Acids, Bases and Salts** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Hardening of plaster of Paris is due to: 1

(a) formation of gypsum **(b)** loss of water
(c) oxidation **(d)** decomposition

2. Which acid is used in car batteries? 1

(a) acetic acid **(b)** nitric acid
(c) hydrochloric acid **(d)** sulphuric acid

3. The salt NH_4Cl in water is: 1

(a) basic **(b)** neutral **(c)** acidic **(d)** amphoteric

4. Which is formed when chlorine is passed through dry slaked lime? 1

(a) plaster of Paris **(b)** bleaching powder **(c)** baking soda **(d)** washing soda

5. Strong acids have: 1

(a) only OH^- **(b)** no ions
(c) low degree of ionisation **(d)** high degree of ionisation

6. A solution turns turmeric paper red. It is: 1

(a) basic **(b)** neutral
(c) a salt solution of NaCl **(d)** acidic

7. In the chlor-alkali process, the electrode at which hydrogen forms is the: 1

(a) cathode **(b)** anode **(c)** both **(d)** neither

8. Which of the following is a weak acid? 1

(a) HCl **(b)** HNO_3 **(c)** H_2SO_4 **(d)** acetic acid

9. Which of the following is the best remedy for a wasp sting (alkaline)? 1

(a) vinegar **(b)** milk of magnesia **(c)** baking soda **(d)** soap solution

10. Which compound gives CO_2 on reaction with an acid? 1

(a) Na_2SO_4 **(b)** $NaHCO_3$ **(c)** NaCl **(d)** NaOH

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does an aqueous solution of Na_2CO_3 turn red litmus blue? 2

12. Why should we not heat gypsum above 373 K to make plaster of Paris? 2

13. Why is sodium hydrogen carbonate used as an antacid? 2

14. What happens when carbon dioxide is passed through lime water and then in excess? Write the first equation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Differentiate between a base and an alkali with examples. 3

16. Explain why baking soda is used in baking and as an antacid. 3

17. Explain the importance of pH in plant and animal life. 3

OR

Explain the reaction of a metal oxide and a non-metal oxide with acid/base. Which is basic and which acidic? 3

18. Explain how the pH changes when dilute HCl is diluted with water repeatedly. 3

OR

Explain how a base is neutralised by a non-metal oxide. Write an example. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how pH helps in everyday life with four examples (digestion, tooth, soil, acid rain). 5

OR

Define acid, base, alkali, indicator and salt with an example of each. 5

20. Describe the chlor-alkali process. Explain the products, equation and their uses. 5

OR

Explain how acids and bases react with metal carbonates, hydrogencarbonates and oxides. Give one equation of each type. 5

END OF QUESTION PAPER — Total: 40 Marks

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